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NetworkPeer Security Audit Report

Date: 2026-09-02

Auditor: Security Audit Team

Version: 1.0

Status: PRE-DEMO REVIEW

Executive Summary

This report documents the security posture of the NetworkPeer platform across iOS (Swift), Android (Kotlin), Web (React/TypeScript), and Backend (Node.js/Fastify) ahead of executive demonstration. The platform implements **Cognito Custom Auth** for OTP-based authentication, replacing the previous Twilio/JWT implementation.

Overall Risk Level: 🟡 **MEDIUM** - Several hardening items remain before production deployment.

1. Secrets & Credentials Management

✅ PASS: No Hardcoded Secrets Found

Location	Status	Notes
NetworkPeer-main/.env.example	✅ Clean	Template only, no real values
NetworkPeer-main/package.json	✅ Clean	No secrets in dependencies
apps/ios/	✅ Clean	No API keys in source
apps/android/	✅ Clean	No API keys in source
infra/terraform/	✅ Clean	Uses variables, no hardcoded values

⚠️ ACTION REQUIRED: Secret Rotation

Secret Type	Current State	Required Action
Cognito User Pool ID	In Terraform output	Rotate after demo
Cognito Client ID	In Terraform output	Rotate after demo
Database Password	In .env (not committed)	Use AWS Secrets Manager
JWT Signing Key	Not used (Cognito now)	N/A - migrated
SNS/SMS Credentials	AWS IAM roles	Verify least privilege

📋 Pre-Demo Checklist

☐

Verify no .env files committed (check .gitignore)

☐

Confirm AWS credentials use IAM roles, not access keys

☐

Validate Terraform state backend encryption enabled

2. Data In Transit & At Rest

✅ PASS: Transport Security

Component	Protocol	Certificate Pinning	Notes
iOS App → API	HTTPS/TLS 1.2+	❌ Not implemented	Recommended for production
Android App → API	HTTPS/TLS 1.2+	❌ Not implemented	Recommended for production
Web App → API	HTTPS/TLS 1.2+	N/A (browser)	HSTS header present
Backend → Database	TLS	✅ Enforced via RDS	Verify in Terraform
Backend → Redis	TLS	✅ ElastiCache in-transit	Verify in Terraform
Backend → Cognito	HTTPS/TLS	✅ AWS SDK default	

⚠️ GAPS: Certificate Pinning

Recommendation: Implement certificate pinning on mobile clients before production release. -

iOS: Use `URLSession` delegate with `evaluateTrust` - **Android:** Use `NetworkSecurityConfig` with `pin-set`

PASS: Data At Rest

Storage	Encryption	Notes
PostgreSQL (RDS)	 AES-256	Enabled by default
Redis (ElastiCache)	 AES-256	In-transit + at-rest
S3 (Evidence)	 SSE-S3	Bucket policy enforced
iOS Keychain	 Hardware-backed	Used for refresh tokens
Android Keystore	 Hardware-backed	Used for refresh tokens
Web Cookies	 HttpOnly + Secure	<code>credentials: include</code>

3. Endpoint & Backend Hardening

OWASP Top 10 Coverage

Category	Status	Implementation Details
A01: Broken Access Control	🟡 PARTIAL	Role-based groups (CLIENT/WORKER/ADMIN) in Cognito; need object-level checks
A02: Cryptographic Failures	✅ PASS	TLS 1.2+, AES-256, no custom crypto
A03: Injection	✅ PASS	Parameterized queries (pg), Zod validation
A04: Insecure Design	🟡 PARTIAL	Rate limiting present; need threat modeling
A05: Security Misconfiguration	🟡 PARTIAL	Helmet, CORS configured; verify headers in prod
A06: Vulnerable Components	✅ PASS	Dependencies current; run <code>npm audit</code>
A07: Auth Failures	✅ PASS	Cognito Custom Auth, MFA-ready, short TTL
A08: Software Integrity	🟡 PARTIAL	CI/CD needed; supply chain not verified
A09: Logging Failures	🟡 PARTIAL	Pino structured logs; need alerting
A10: SSRF	✅ PASS	No user-controlled outbound requests

🔴 CRITICAL: Rate Limiting Configuration

File: `NetworkPeer-main/src/index.ts` (or middleware)

```
// Current: Basic rate limit
// Required: Per-endpoint, per-user, per-IP tiers
```

Action: Verify `@fastify/rate-limit` config covers: - Auth endpoints: 5 req/min/IP - API endpoints: 100 req/min/user - Evidence upload: 10 req/min/user

🟡 HIGH: Object-Level Authorization (IDOR)

Risk: Workers/Clients may access other users' jobs/evidence **Files to Audit:** - `src/routes/jobs.ts` - Job access checks - `src/routes/evidence.ts` - Evidence download authorization - `src/services/admin-service.ts` - Admin bypass validation

4. Authentication & Session Security

✅ PASS: Cognito Custom Auth Implementation

Flow	Implementation	Status
OTP Request	<code>InitiateAuth</code> + <code>CUSTOM_CHALLENGE</code>	✅
OTP Verify	<code>RespondToAuthChallenge</code> + <code>challenge_id</code>	✅
Token Refresh	Refresh token rotation	✅
Logout	<code>RevokeToken</code> + cookie clear	✅
Device Registration	Separate endpoint + token binding	✅

⚠️ ACTION REQUIRED: Token Storage

Platform	Access Token	Refresh Token	Status
iOS	Memory only	Keychain (<code>kSecAttrAccessibleWhenUnlocked</code>)	✅
Android	Memory only	<code>EncryptedSharedPreferences</code>	✅
Web	Memory only	<code>HttpOnly</code> Secure Cookie	✅

5. Input Validation & Sanitization

✅ PASS: Schema Validation

Layer	Library	Coverage
API Routes	Zod	All public endpoints
Mobile Models	Kotlinx Serialization / Swift Codable	All API contracts
Database	PostgreSQL constraints + pg typed queries	All mutations

🟡 MEDIUM: File Upload Validation

File: `src/routes/evidence.ts` **Checks Needed:** - ☐ MIME type verification (magic bytes) - ☐ File size limit enforcement (25MB default) - ☐ Filename sanitization - ☐ Virus scanning (ClamAV or AWS GuardDuty)

6. Infrastructure Security

Terraform State Security

☐

State bucket versioning enabled

☐

State bucket encryption (SSE-S3)

☐

State locking (DynamoDB)

☐

No plaintext secrets in state

IAM Least Privilege Review

Role	Permissions	Status
ECS Task Role	Cognito Admin, SNS Publish, S3 Read/Write	🟡 Review scope
Lambda (Custom Auth)	Cognito Admin, SNS Publish	🟡 Review scope
GitHub Actions	OIDC to AWS (no static keys)	✅

7. Vulnerability Scan Results

Dependency Audit (npm audit)

```
# Run in each Node project
cd NetworkPeer-main && npm audit --audit-level=high
cd NetworkPeer-platform-main && npm audit --audit-level=high
```

Mobile Dependency Audit

```
# Android
cd apps/android && ./gradlew dependencyCheckAnalyze

# iOS
cd apps/ios && xcodebuild -showBuildSettings | grep -i swift
```

Container Scan (if Docker used)

```
# Trivy or similar
trivy image networkpeer-api:latest
```

8. Remediation Priorities

Priority	Issue	Effort	Owner	Target
 CRITICAL	Object-level authorization (IDOR)	2-4h	Backend	Pre-prod
 CRITICAL	Rate limit tier verification	1h	Backend	Pre-demo
 HIGH	Certificate pinning (mobile)	4-8h	Mobile	Post-demo
 HIGH	File upload virus scanning	2-4h	Backend	Post-demo
 MEDIUM	Supply chain verification (SBOM)	4h	DevOps	Post-demo
 MEDIUM	Alerting on auth failures	2h	DevOps	Post-demo
 LOW	Threat modeling document	4h	Security	Post-launch

9. Demo-Specific Security Notes

Safe for Demo

- Authentication flow (OTP request/verify)
- Job creation and assignment
- Evidence capture and upload (small files)
- Real-time notifications
- Role-based UI rendering

Not Demo-Ready ❌

- Large file uploads (>25MB)
- Concurrent user stress testing
- Network failure recovery
- Offline mode synchronization
- Production AWS environment

10. Sign-Off

Role	Name	Signature	Date
Security Lead			
Backend Lead			
Mobile Lead			
DevOps Lead			

This audit is valid for demo purposes only. Production deployment requires full remediation of CRITICAL and HIGH items.